AI for Software Engineers Course

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DAY – 1 : Core AI + Machine Learning Foundations (Engineer-Oriented)

- *AI landscape for engineers*
- *How ML systems actually work*
- *Core ML types (practical framing)*
- *Neural networks intuition*
- *Homeworks (2)*
- *Outcome: Understand ML as a software system.*

DAY – 2 : LLMs, Generative AI, and Modern AI Stack

- *What changed: from ML to LLMs*
- *Transformers (engineer-level explanation)*
- *How LLMs work in production*
- *Embeddings (VERY important for engineers)*
- *Prompting (practical engineering skill)*
- *RAG (Retrieval Augmented Generation), Ontology, LLM wiki*
- *SLM Distillation*
- *Homeworks(2)*
- *Outcome: Understand how modern AI products are actually built.*

DAY – 3 : Building AI Systems + Production + Real Use Cases

- *AI in real software architecture*
- *LLM application patterns*
- *AI agents (modern topic)*
- *MLOps & LLMOps (important for engineers)*
- *Cost, latency, and scaling*
- *Security and risks (engineering angle)*
- *Homeworks (2)*
- *Outcome: Design and prototype real AI-powered systems.*

Extra Topics

- *Fine-tuning vs RAG tradeoff*
- *Open-source models (Llama, Mistral)*
- *Vector databases deep dive*
- *Multimodal models (text + image)*
- *Agent frameworks (LangChain, LangGraph, LlamaIndex)*